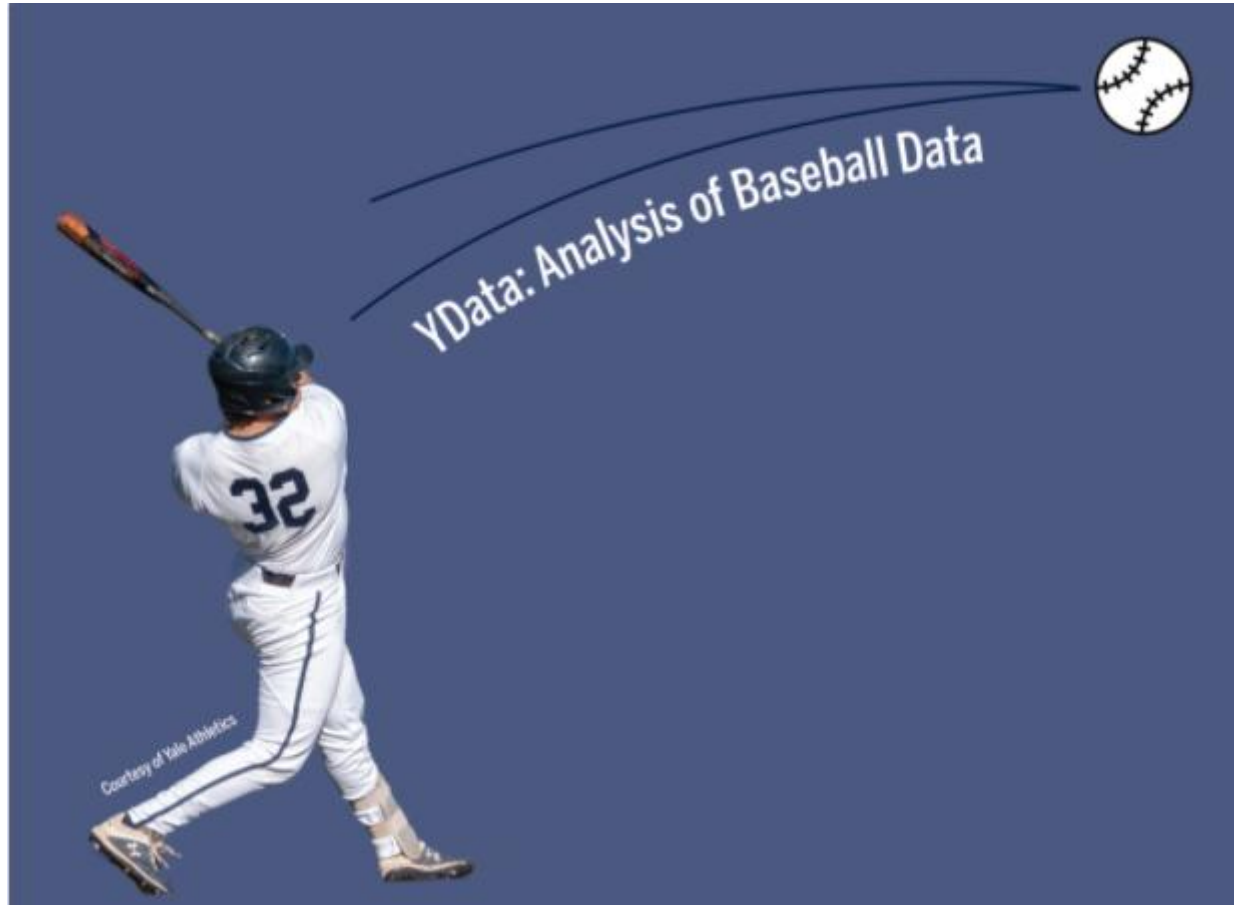


S&DS 1730



Class 4: Assessing what are impressive baseball statistics

Overview

Are any of these statistics impressive?

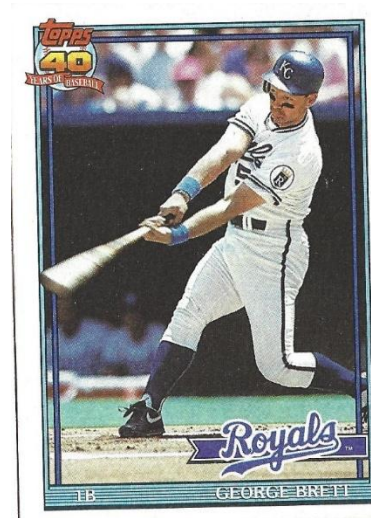
Brief discussion of Astrobball

Very quick review of pandas

Assessing "impressive" statistics using percentiles, boxplots, histograms, and other forms of data visualization

- Visualizing data using matplotlib and seaborn

Examining how statistics values have changed over time, and comparing players across eras

A Topps 540 baseball card featuring George Brett. The card includes a biographical section at the top and a detailed batting record table below.

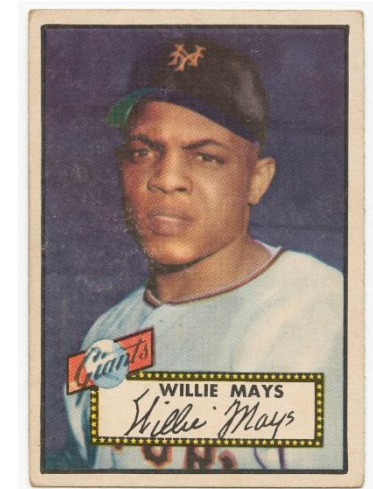
HT: 6'0" WT: 200 BATS: LEFT THROWS: RIGHT DRAFT: ROYALS #2-JUNE, 1971
ACQ: VIA DRAFT BORN: 5-15-53, OLEN DALE, WEST VIRGINIA HOME: RANCHO MIRAGE, CALIF.

GEORGE BRETT • 1B

COMPLETE MAJOR LEAGUE BATTING RECORD (LEAGUE LEADER IN ITALICS, TIE +)

YR	CLUB	G	AB	R	H	2B	3B	HR	RBI	SB	SLG	BB	SO	Avg
73	ROYALS	13	40	2	5	2	0	0	0	0	.175	0	5	.125
74	ROYALS	133	457	40	129	21	5	2	47	9	.363	21	38	.282
75	ROYALS	159	634	84	195	35	7	11	89	13	.456	46	49	.308
76	ROYALS	158	645	84	215	34	7	7	87	21	.462	48	38	.337
77	ROYALS	139	564	109	176	32	13	22	88	14	.532	55	24	.312
78	ROYALS	128	510	70	150	45	8	9	62	23	.467	39	35	.294
79	ROYALS	154	645	119	212	42	20	23	107	17	.583	51	36	.329
80	ROYALS	117	440	87	175	33	9	24	118	15	.664	38	22	.390
81	ROYALS	86	347	42	109	27	7	8	43	14	.484	27	23	.314
82	ROYALS	144	552	101	166	32	5	21	82	6	.505	71	51	.301
83	ROYALS	122	484	80	144	38	2	25	93	0	.563	37	38	.316
84	ROYALS	104	377	42	107	21	5	15	89	0	.459	38	37	.284
85	ROYALS	155	550	108	184	36	5	30	112	9	.595	103	49	.335
86	ROYALS	124	441	70	128	28	4	16	73	1	.481	80	45	.280
87	ROYALS	115	427	71	124	18	2	22	78	6	.486	72	47	.287
88	ROYALS	157	588	90	180	42	3	24	103	14	.589	82	51	.306
89	ROYALS	124	457	87	129	25	3	12	80	14	.431	39	47	.282
90	ROYALS	142	544	82	179	45	7	14	87	9	.515	56	63	.329
MAJ. LEA. TOTALS		2278	8682	1382	2707	558	127	281	1388	184	.502	884	686	.311

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Announcement: Lab 3

Lab 3 has been released. It is due on
Sunday February 8th

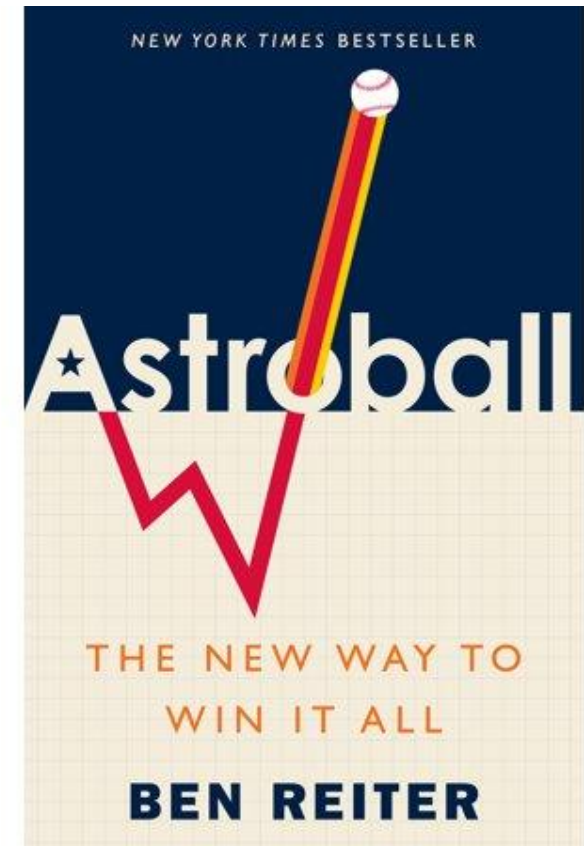
How is did lab 1 and 2 go?



Discussion on chapter 2 of Astrobball

Let's discuss the chapter 2 for 3 minutes in small groups of 3-4 people and then have a larger conversation as a group

- E.g., Discuss your quote and reaction to chapter 1

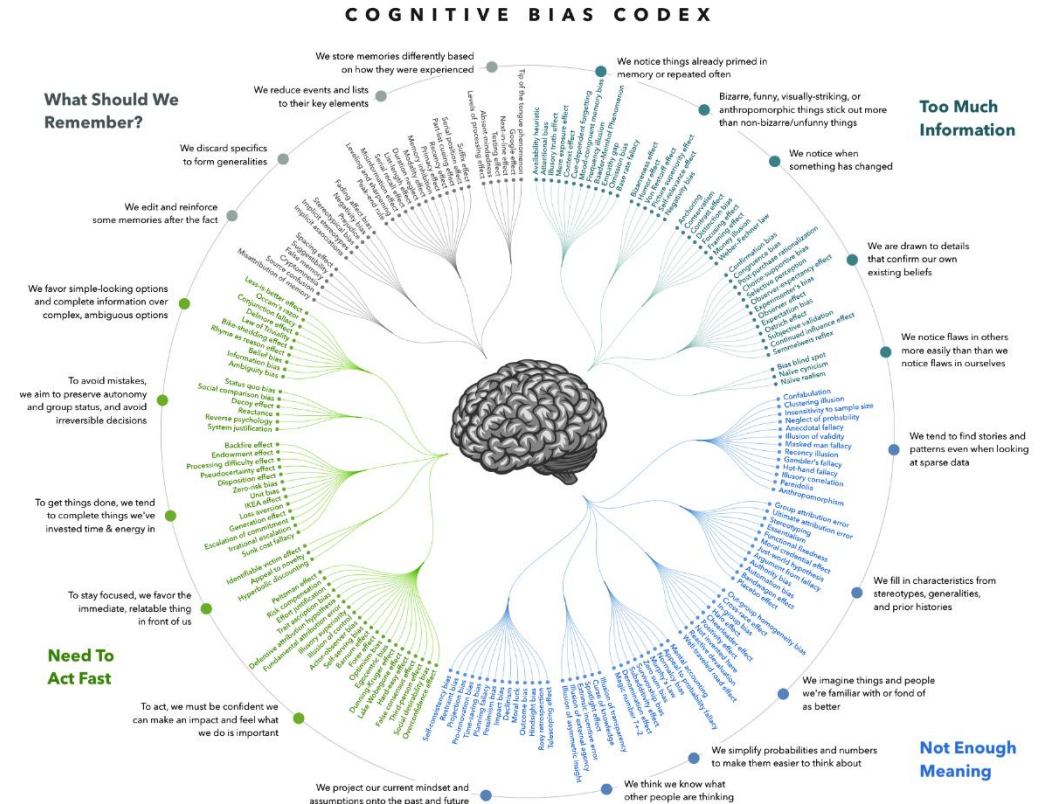


Astroball: making the best decisions

Using linear regression to combining performance statistics and scouting reports

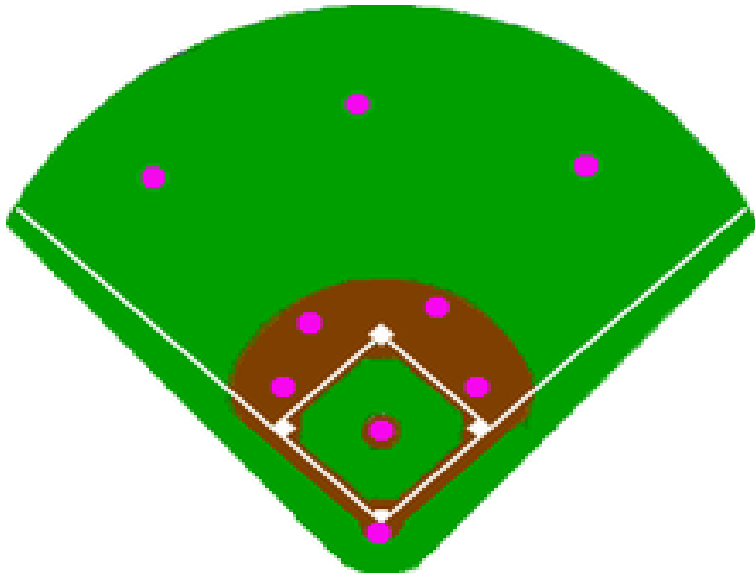
Kahneman and Tversky

- Behavioral economics vs. assuming rational agent
- Cognitive biases

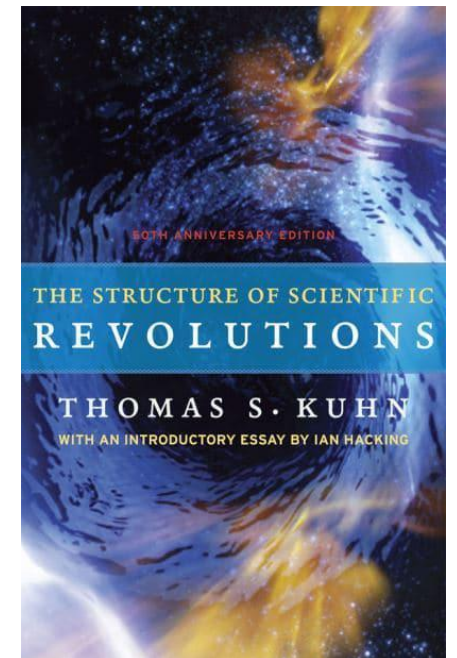
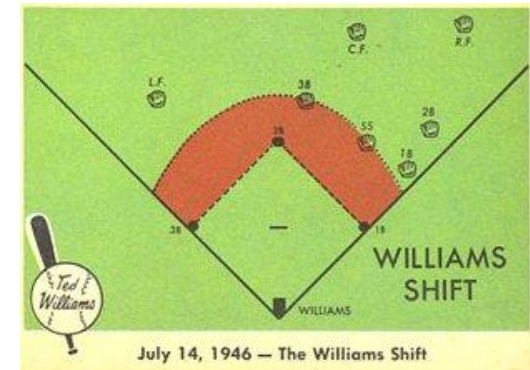
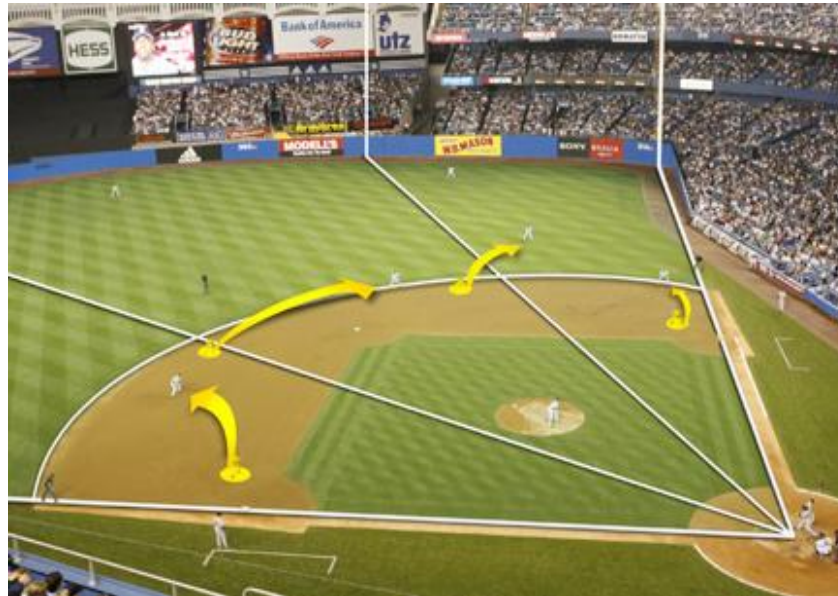


Astroball: infield shifts

Normal positioning



Defensive shift



Astrobball: the butt slide

Deadspin from every angle

MLB analysis

Quick review: pandas DataFrames



Quick review: pandas DataFrames

Pandas DataFrames hold Table data

	playerID	yearID	stint	teamID	lgID	G
0	aardsda01	2004	1	SFN	NL	11
1	aardsda01	2006	1	CHN	NL	45
2	aardsda01	2007	1	CHA	AL	25
3	aardsda01	2008	1	BOS	AL	47

Selecting columns:

- `my_df[["col1", "col2"]]` # getting multiple columns using a list

Extracting rows:

- `my_df.iloc[0]` # getting a row by number
- `my_df.loc["index_name"]` # getting a row by Index value
- `my_df[my_df["col_name"] == 7]` # getting rows using a Boolean mask
- `my_df.query("col_name == 7")` # getting rows using the query function

Quick review: pandas DataFrames

Sorting rows of a DataFrame

```
my_df.sort_values("col_name", ascending = False)  # sort from largest to smallest
```

Adding a new:

```
my_df["new_col"] = values_array
```

Renaming a column:

```
rename_dictionary = {"old_col_name": "new_col_name"}  
my_df.rename(columns = rename_dictionary )
```

Creating aggregate statistics by group

We can get statistics separately by group:

```
baseball.groupby("Year").agg("sum")
```

```
baseball.groupby("Year")["HR"].sum()
```

DataFrame

Col 1	Col 2
A	1
B	2
C	3
C	4
B	2
B	5
A	3

```
my_df.groupby("group_col_name").agg(  
    new_col1 = ('col_name', 'statistic_name1'),  
    new_col2 = ('col_name', 'statistic_name2'),  
    new_col3 = ('col_name', 'statistic_name3')  
)
```

```
batting.groupby("teamID").agg(  
    max_HR = ("HR", "max"),  
    first_player = ("playerID", "min")  
)
```

Review: Merging/Joining DataFrames

The DataFrames in the Lahman database can be connected through different ID columns

- For example, we can relate the Batting and People DataFrames using the playerID column

Batting DataFrame

	playerID	yearID	stint	teamID	lgID	G	G_batting	AB	R	H
1	aardsda01	2004	1	SFN	NL	11	11	0	0	0
2	aardsda01	2006	1	CHN	NL	45	43	2	0	0
3	aardsda01	2007	1	CHA	AL	25	2	0	0	0
4	aardsda01	2008	1	BOS	AL	47	5	1	0	0
5	aardsda01	2009	1	SEA	AL	73	3	0	0	0
6	aardsda01	2010	1	SEA	AL	53	4	0	0	0
7	aardsda01	2012	1	NYA	AL	1	NA	NA	NA	NA
8	aaronha01	1954	1	ML1	NL	122	122	468	58	131

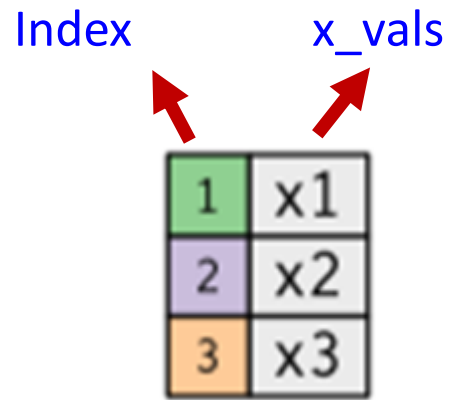
People DataFrame

	playerID	nameFirst	nameLast	birthYear	birthMonth
1	aardsda01	David	Aardsma	1981	12
2	aaronha01	Hank	Aaron	1934	2
3	aaronto01	Tommie	Aaron	1939	8
4	aasedo01	Don	Aase	1954	9
5	abadan01	Andy	Abad	1972	8

Joining

Suppose we have two DataFrames (or Series) called **x_df** and **y_df**

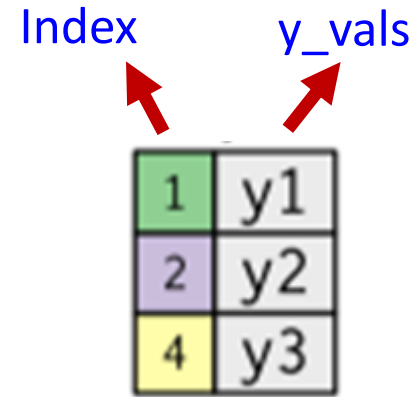
- **x_df** have one column called **x_vals**
- **y_df** has one column called **y_vals**



A diagram of a DataFrame with two columns. The first column is labeled 'Index' with a red arrow pointing to it. The second column is labeled 'x_vals' with a red arrow pointing to it. The DataFrame contains three rows: the first row has index 1 and value x1; the second row has index 2 and value x2; the third row has index 3 and value x3.

Index	x_vals
1	x1
2	x2
3	x3

DataFrame: x_df



A diagram of a DataFrame with two columns. The first column is labeled 'Index' with a red arrow pointing to it. The second column is labeled 'y_vals' with a red arrow pointing to it. The DataFrame contains three rows: the first row has index 1 and value y1; the second row has index 2 and value y2; the third row has index 4 and value y3.

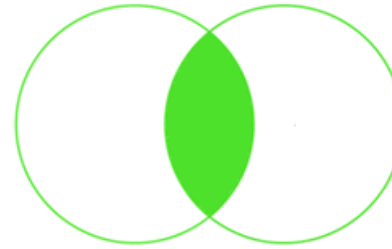
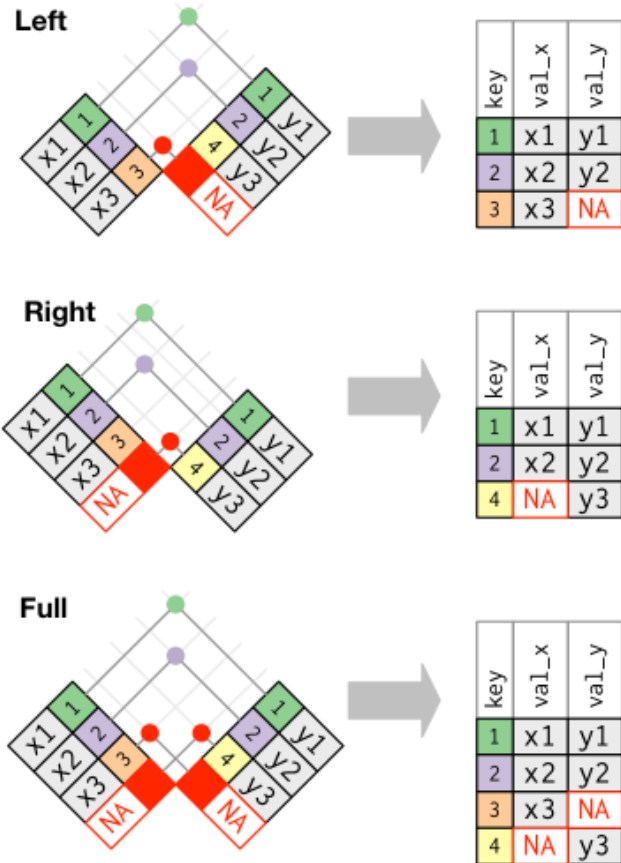
Index	y_vals
1	y1
2	y2
4	y3

DataFrame: y_df

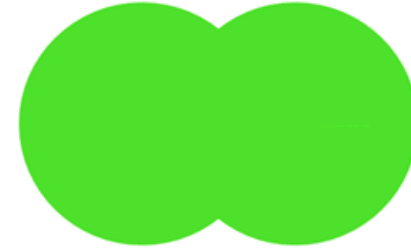
We can join these two DataFrames into a single DataFrame by aligning rows with the same Index value using the general syntax: **x_df.join(y_df)**

- i.e., the new joined data frame will have two columns: **x_vals**, and **y_vals**

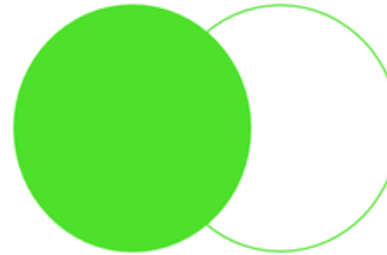
Joining



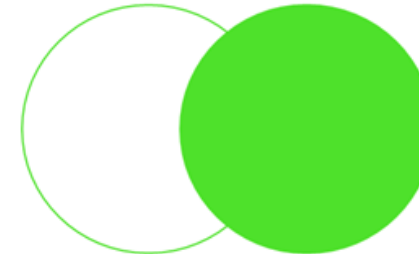
INNER JOIN



FULL OUTER JOIN



LEFT JOIN



RIGHT JOIN

`x_df.join(y_df, how = "left")` # based on Index

`x_df.merge(y_df, how = "inner", left_on = "x_col", right_on = "y_col")` # based on columns

Warm-up exercises

Let's do a few quick warmup exercises to practice manipulating DataFrames in pandas



Percentiles, boxplots, and histograms

What are "impressive" value for particular
baseball statistics?

Motivation

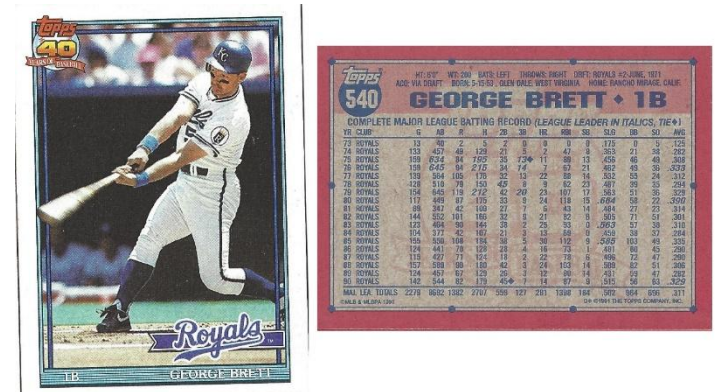
We are now able to generate the statistics on any baseball card with the function we wrote on lab 2: `get_baseball_card()`

- E.g., `get_baseball_card("George", "Brett", 1990)`

Are any of these statistics impressive?

Questions:

- How can we tell if these statistics are impressive?
- Is it fair to compare statistics across eras?
- Are new sabermetric statistics, such as wOBA better?
 - What is the best statistic to use to assess a player?



What are “impressive” statistic values?

How could we determine what a “good” value for a statistic is?

- i.e., how many home runs would need to be hit to determine if a player is “good” at hitting home runs?

One method to determine what a “good” statistic is, would be a value that is say greater than 90% of baseball players

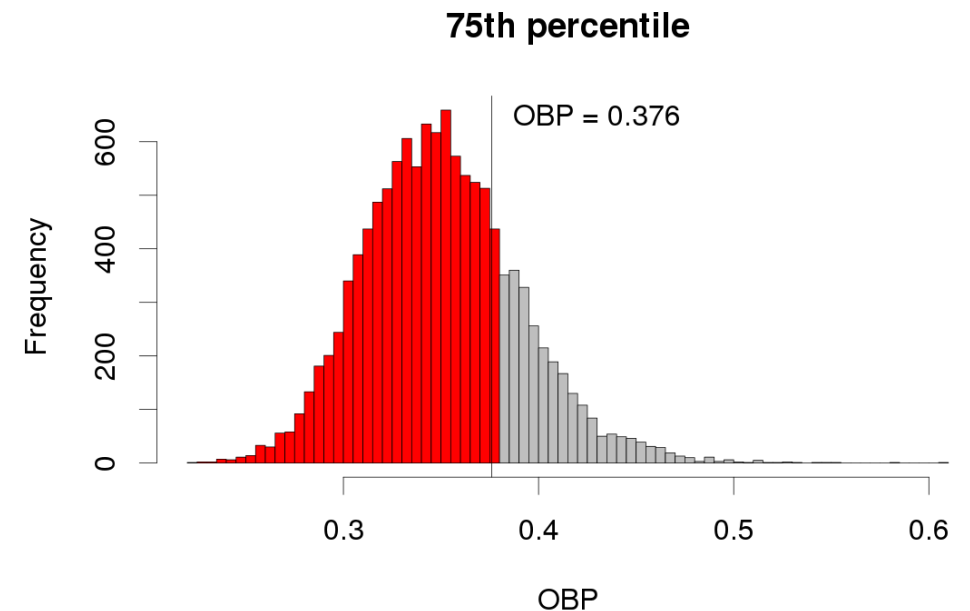
Percentiles

The **p^{th} percentile** is the value **v** of a quantitative variable such that p percent of the data is less than **v**

We can use the `my_df.quantile()` get a percentile value:

get the 90th percentile

```
my_df.quantile(.9)
```



https://emeyers.shinyapps.io/baseball_stat_percentiles/

What are "impressive" statistic values?

In lab 3 you will use percentiles to find what "impressive" statistic values are many statistics including:

- Home runs (HR)
- Batting Average (AVG)
- Slugging Percentage (SLG)
- On-base percentage (OBP)
- On-base plus Slugging (OPS)

[illegible]

Five Number Summary

We can summarize a sample of data using a five number summary

Five Number Summary = (min, Q_1 , median, Q_3 , max)

Q_1 = 25th percentile (also called 1st quartile)

Q_3 = 75th percentile (also called 3rd quartile)

Roughly divides the data into fourths

```
my_df.quantile([0, .25, .5, .75, 1])
```

Range and Interquartile Range

Range = maximum – minimum

Interquartile range (IQR) = $Q_3 - Q_1$

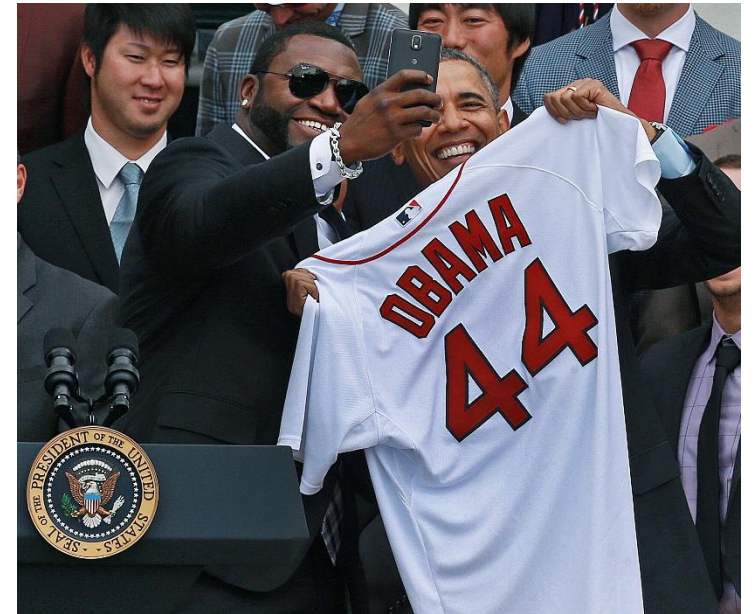
- Range where the middle 50% of the data lies

Detecting of outliers

As a rule of thumb, we call a data value an **outlier** if it is:

Smaller than: $Q_1 - 1.5 * IQR$

Larger than: $Q_3 + 1.5 * IQR$



Are there any outlier years in David Ortiz home run numbers?

1. Five Number Summary: (23, 28.5, 32, 36, 54)

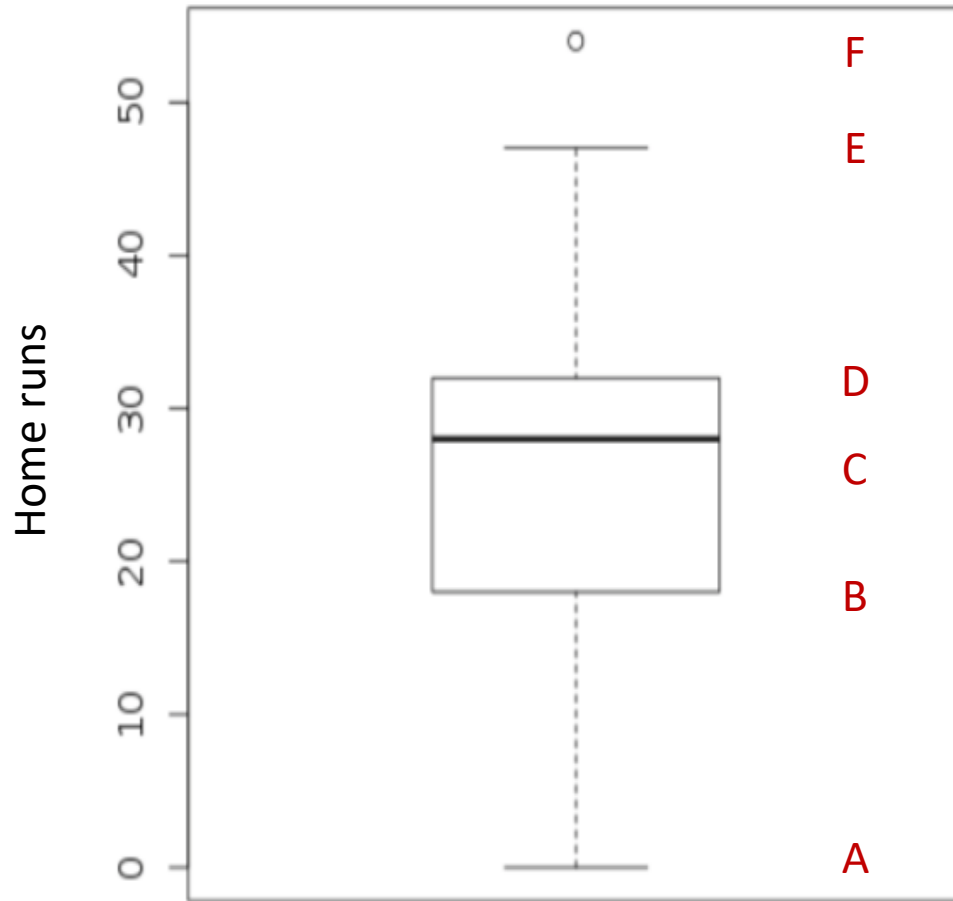
2. Interquartile range (IQR) = 7.5

Box plots

A **box plot** is a graphical display of the 5 number summary and consists of:

1. Drawing a box from Q_1 to Q_3
2. Dividing the box with a line drawn at the median
3. Draw a line from each quartile to the most extreme data value that is not and outlier
4. Draw a dot/asterisk for each outlier data point.

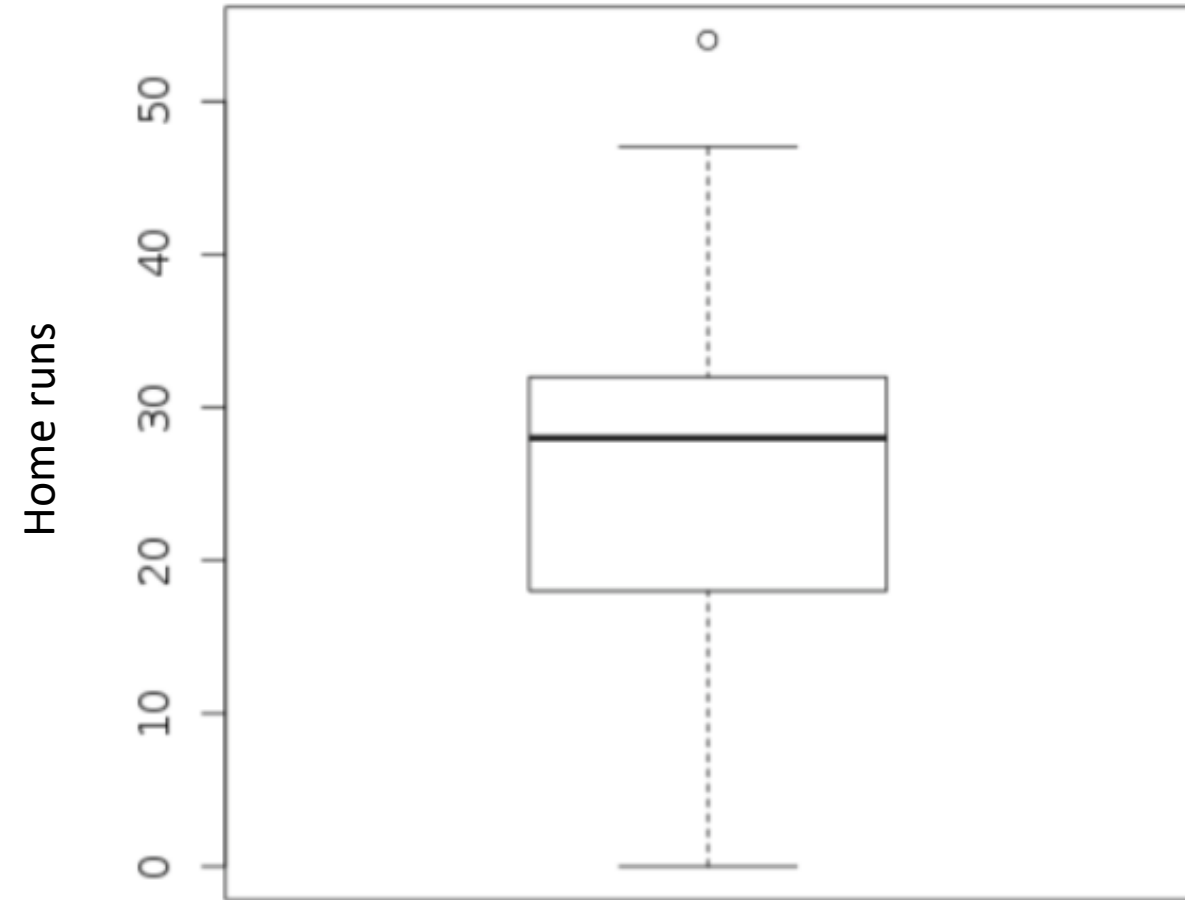
Box plot of David Ortiz home runs



What is:

- Q1?
- Q3?
- The median?
- Most extreme values that are not outliers
- Outliers

Box plot of David Ortiz home runs



We can create boxplots in Python using matplotlib's `boxplot()` function

```
import matplotlib.pyplot as plt
```

```
plt.boxplot(data)
```

Who is better?

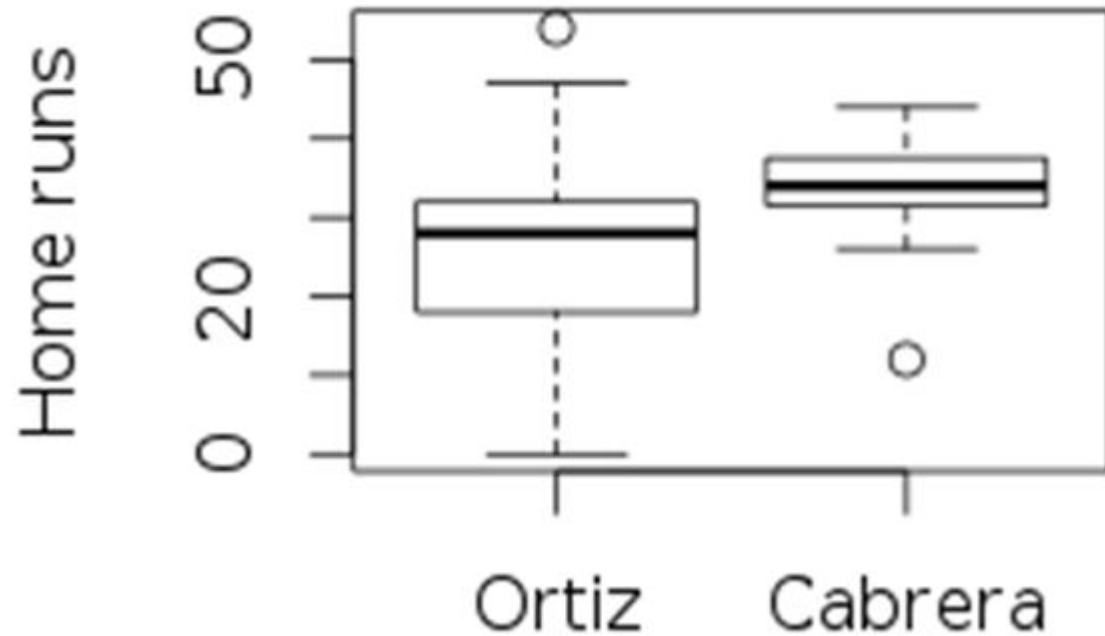


Miguel Cabrera:
HR in 2014 = 25



David Ortiz:
HR in 2014 = 35

Comparing players with side-by-side box plots



```
plt.boxplot([data1, data2],  
            tick_labels = ["A", "B"])
```

How would you describe the differences between these two players in terms of HRs?

Who is better?

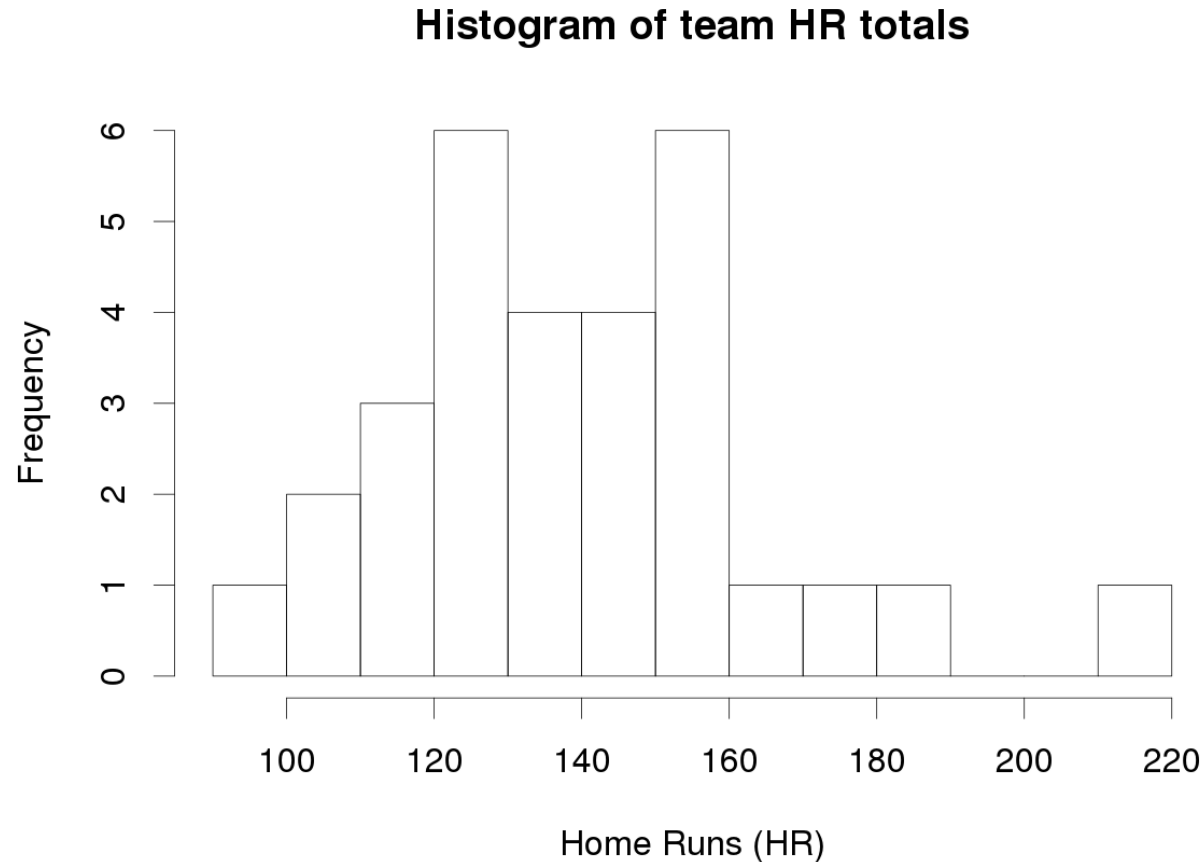
Visualizing the 'shape' of how data is distributed

Boxplots can give us a sense of some key statistics about our data

There are other methods that can give us a better picture of the shape of how all the data is distributed

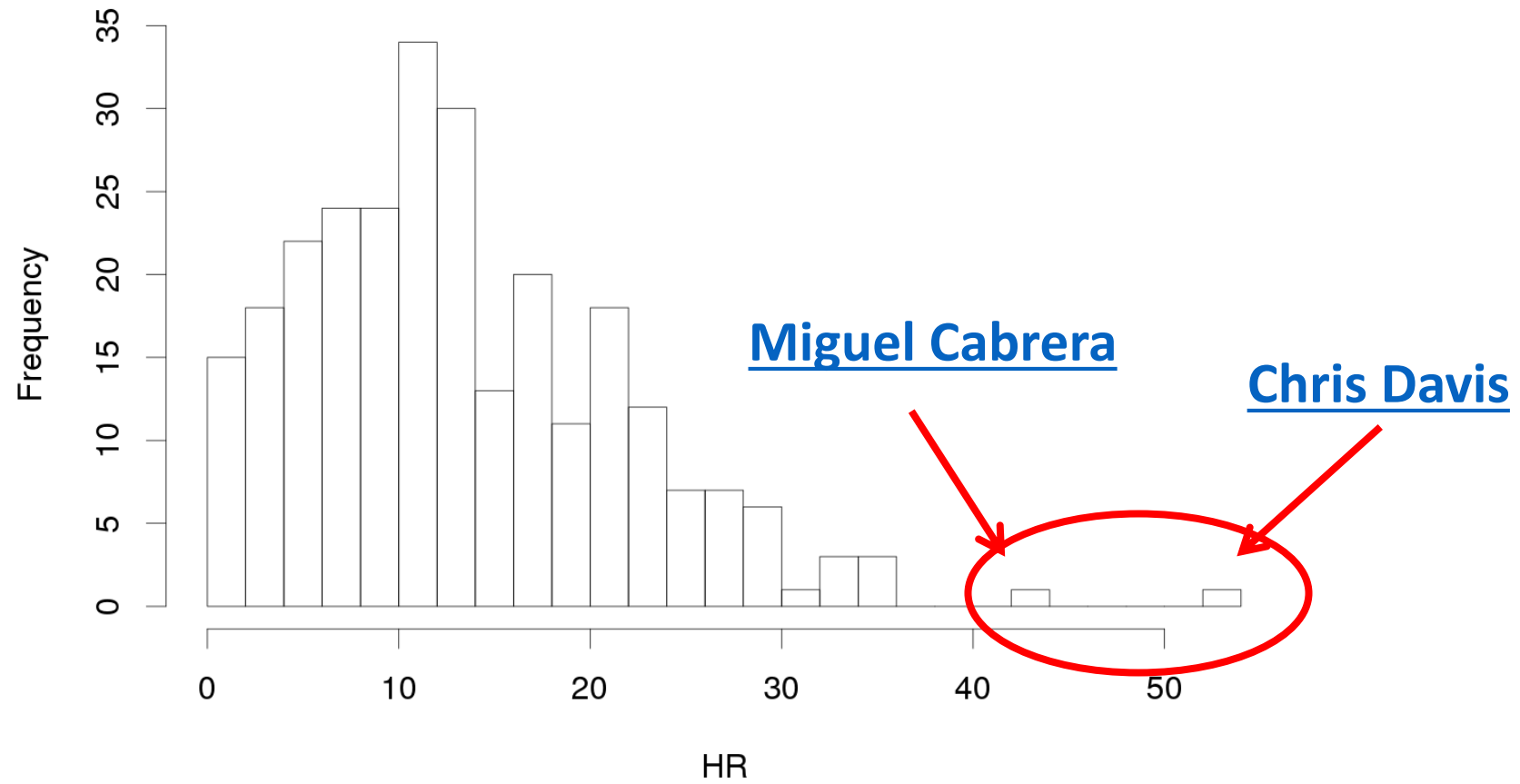
Histograms

Another related way to get a sense of the shape of a distribution is to use a **histogram**



`plt.hist(data1)`

Histogram of HRs for 2013 players with over 300 PA



Observations about the distribution?

Summary of relevant functions

Percentiles

```
my_df.quantile([.25 .75])
```

Data visualization

```
import matplotlib.pyplot as plt
```

```
plt.boxplot([data1, data2], tick_labels = ["A", "B"])
```

```
plt.hist(data1)
```

Let's try it in Python!



Visualizing data with seaborn

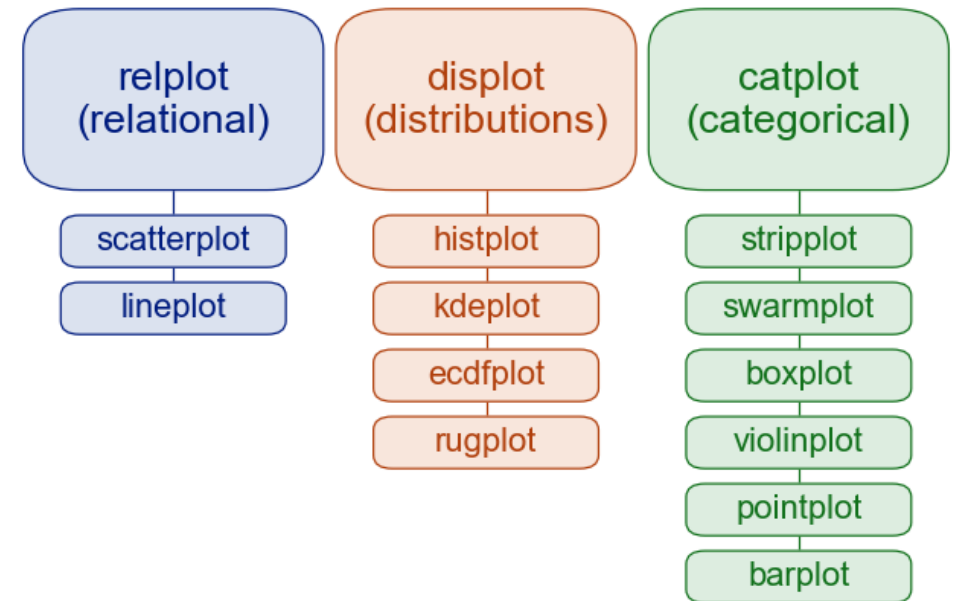
Seaborn figure level plots

Figure level plots are grouped based on the types of variables being plotted

In particular, there are plots for:

1. Two quantitative variables
 - `sns.relplot()`
2. A single quantitative variable
 - `sns.displot()`
3. Quantitative variable compared across different categorical levels
 - `sns.catplot()`

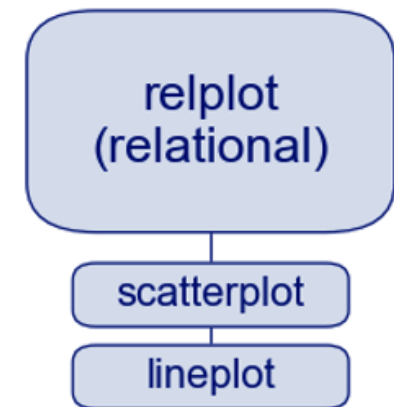
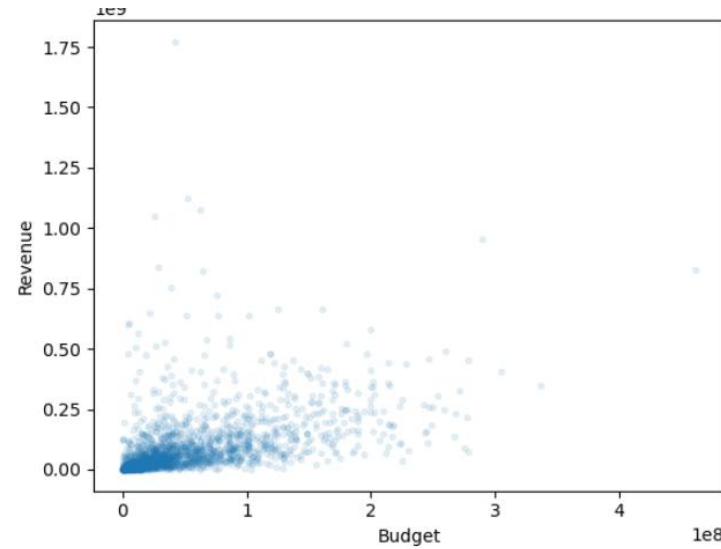
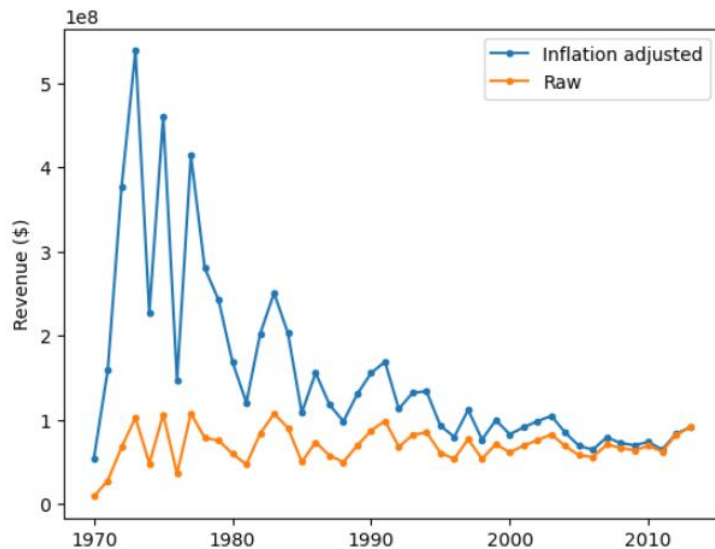
Figure level plots



Plots for two quantitative variable

What types of plots show the relationships between two quantitative variable?

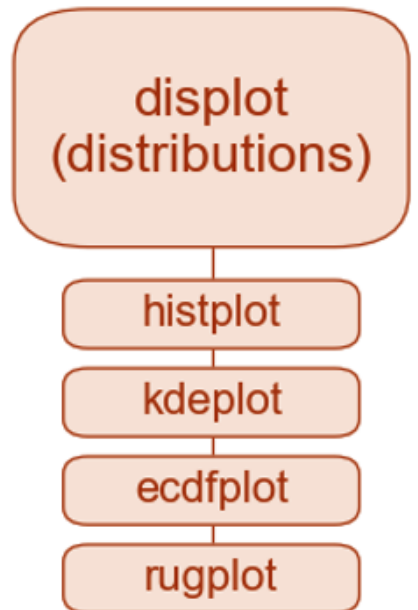
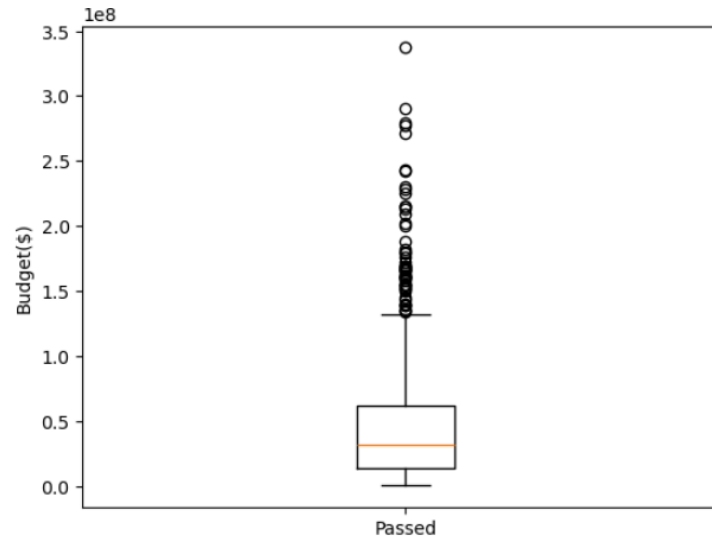
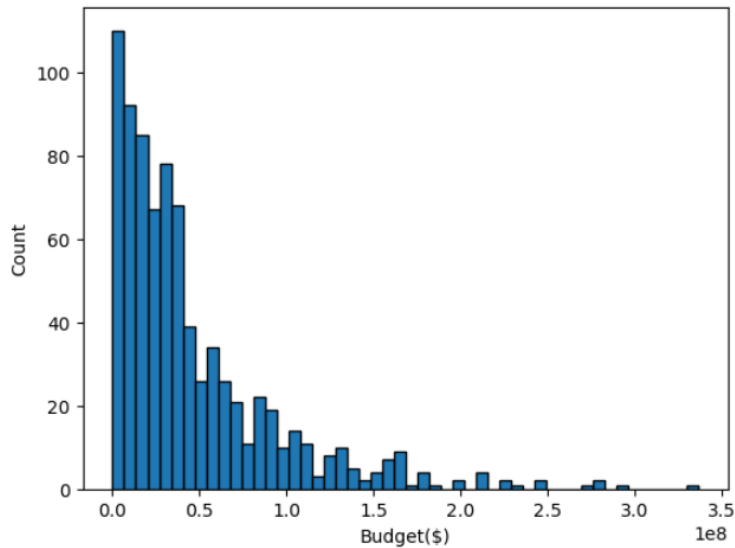
- Line plots and scatter plots!



Plots for a single quantitative variable

What types of plots visualize a single quantitative variable?

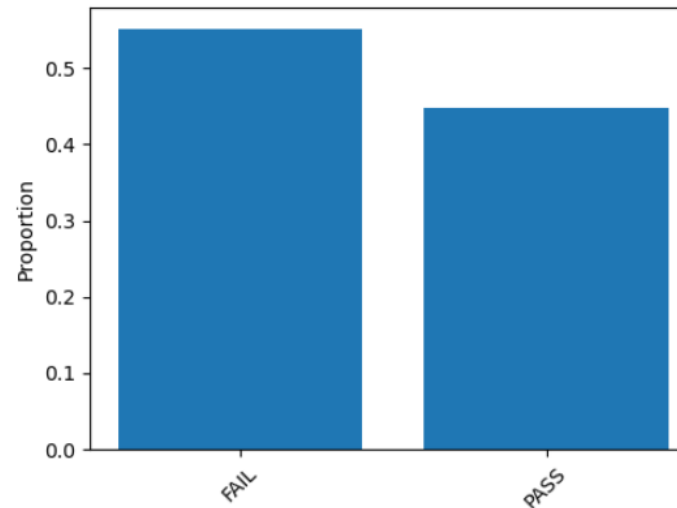
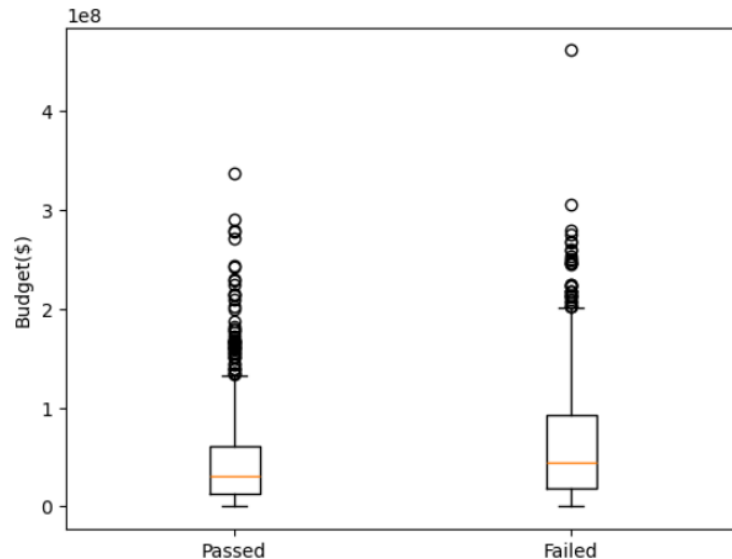
- Histograms and boxplots



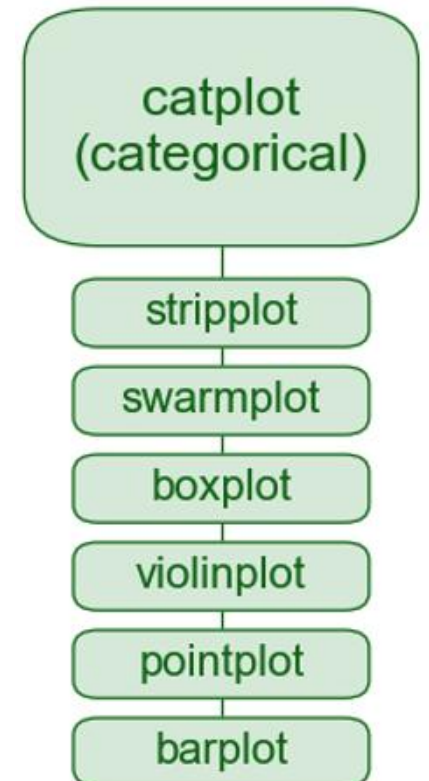
Plots for quantitative data comparing across different categorical levels

What types of plots compare quantitative data at different levels of a categorical variable?

- Side-by-side boxplots, barplots (sort of)



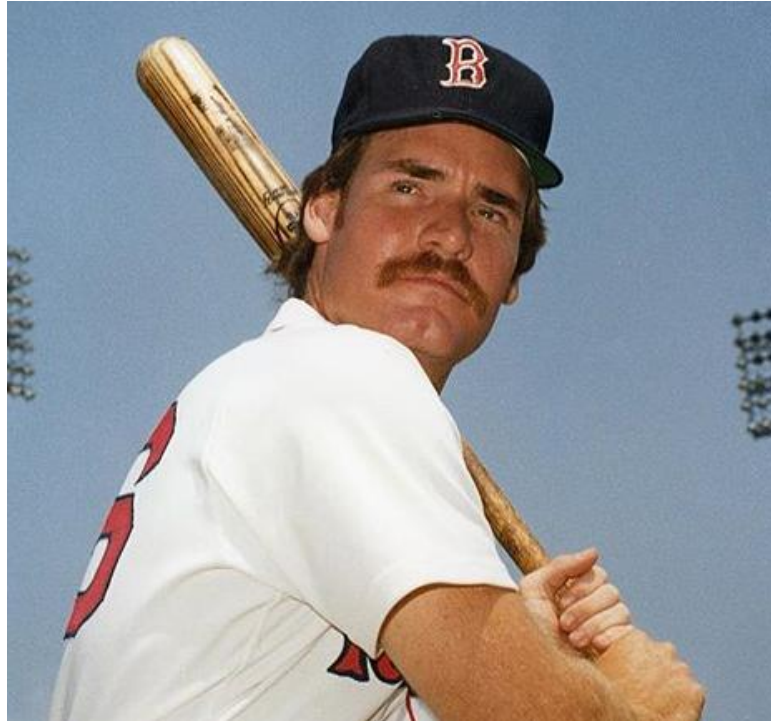
Let's explore this in Jupyter!



Comparing players from different eras

Let's compare two more players

1985



Wade Boggs: BA = .368

Career best seasons

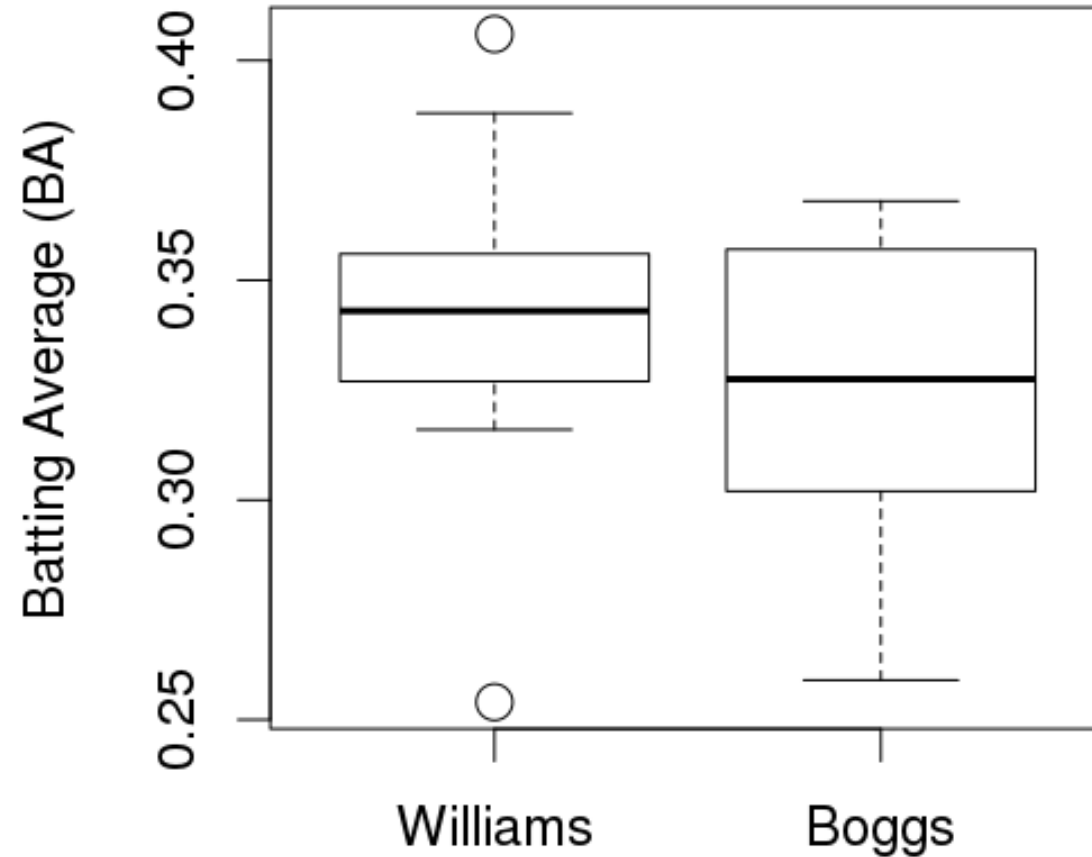
1941



Ted Williams: BA = .406

Who is better?

Who is best here?



Is Ted Williams better than Wade Boggs?

Ted Williams hit .406 in 1941
23 people hit over .400 before him in MLB
but no one in the MLB has since...



Examining how the game has changed

Let's figure out how to plot the MLB leader in batting average each year to see how the game has changed

- i.e., what is the highest BA every year since 1871

Let's try it in Python!

Comparing players across eras

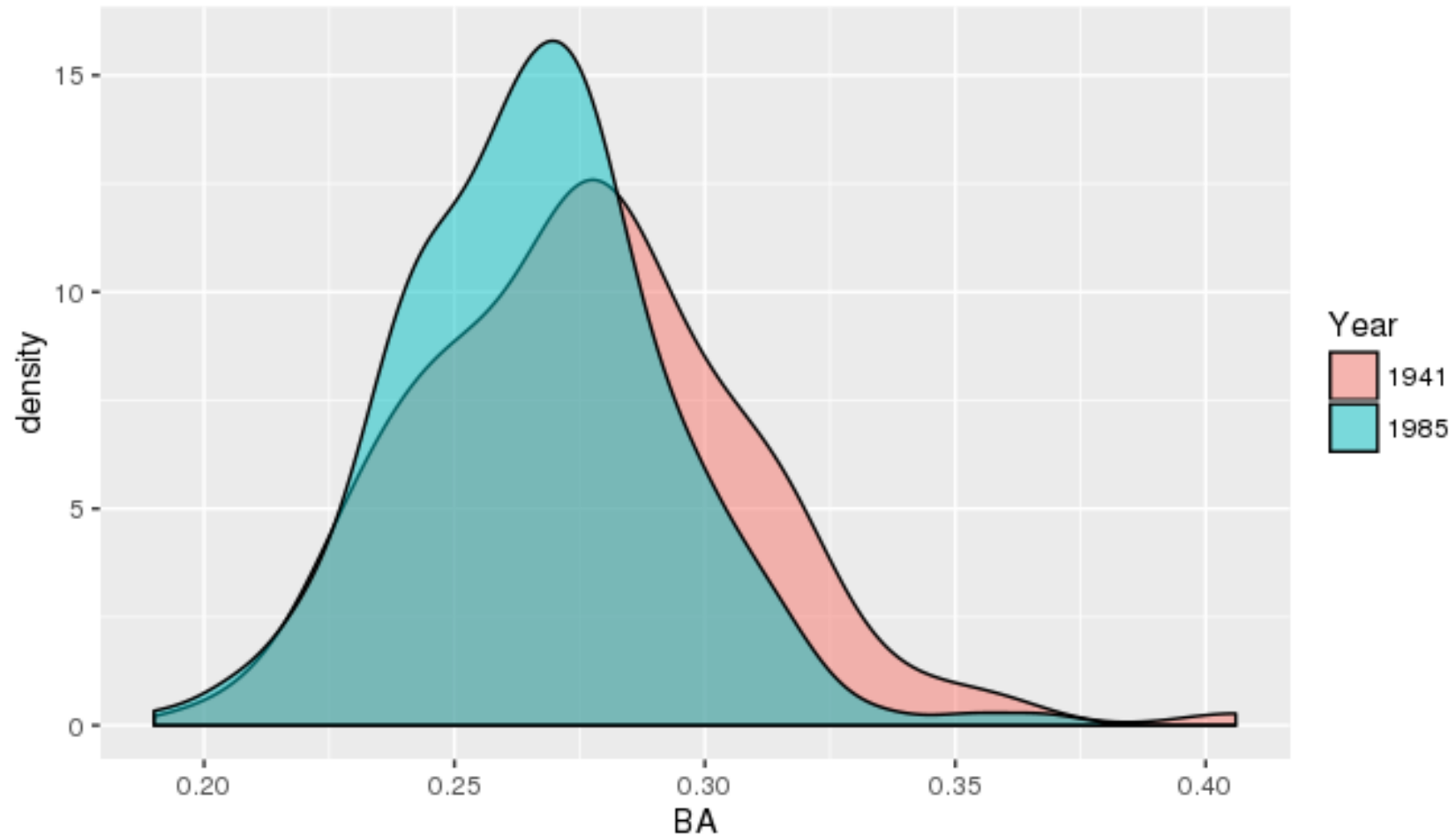
Comparing players across time periods

Problem: baseball has changed from 1871 to now

We can't simply compare statistics to judge how great a baseball player is when comparing across decades

Useful to judge the 'greatness' of players relative to their peers

Density of batting average 1941 vs. 1985



Do the batting averages look similar in these years?

z-scores

The z-scores tells how many standard deviations a value x is from the mean ($\bar{x}$)

- z-scores are independent of the units of measurement

$$z_i = \frac{x_i - \bar{x}}{s}$$

When comparing players across years, we will use the mean ($\bar{x}$), and standard deviation (s) from each year separately

- I.e., we will create separate z-scores for each year

This will give a measure of player performance relative to their peers in the same era

Comparing Ted and Wade to their peers

$$z_i = \frac{x_i - \bar{x}}{s}$$

In 1941:

- Mean batting average was: .276
- Standard deviation in batting average was: .033
- Ted William's batting average was: .406

In 1985:

- Mean batting average was: .266
- Standard deviation in batting average was: .027
- Wade Bogg's batting average was: .368

Calculate z-scores for Ted and Wade's batting averages.

Who was better relative to their peers?

Comparing Ted and Wade's full career

Use Python to get z-scores from Ted and Wade for all years in their careers

Create side-by-side boxplots comparing their z-score batting averages



Let's try it in Python!

For next class

1. Read chapter 3 of Astrobball
2. Lab 3: due 11pm on Sunday

Next class guest: Sean Ahmed!

